

Yeonjun Lee

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EDUCATION

Korea International School Seoul, Republic of Korea

08/2019 - Present

HONORS & ACHIEVEMENTS

- **Korea Code Fair (Top 70):** 2025
 - Selected as one of the top 70 student developers nationwide in a prestigious government-sponsored contest serving as a preliminary for the International Science and Engineering Fair (ISEF).
 - Recognized for technical innovation and social utility among a highly competitive pool of high school applicants across South Korea.
- **Korea Science and Engineering Fair (KSEF)** 2025
 - Selected to compete in South Korea's largest and most prestigious national science research competition.
 - Presented original research in the social science division among the nation's top high school student researchers.
 - Awarded honorable mention
- **Principal's Honor Roll** 2023 – Present
 - Achieved distinction for **five consecutive semesters** (Freshman through Junior year) by maintaining a GPA above 3.66.
 - Recognizes consistent academic excellence and dedication to a rigorous high school curriculum

EXTRACURRICULAR ACTIVITIES

Summer Program

UChicago Summer Session: Understanding AI

July 2025

- Selected as one of **30 students** for this intensive 3-week course, I engaged in high-level discourse on the intersection of technical AI development and socio-ethical responsibility.
- Engineered and designed a custom chatbot, dissecting the transformer architecture to identify technical constraints and vulnerabilities in current generative models.
- Conducted a **Socio-Ethical Audit** of AI systems, developing a strategic framework for **Responsible AI** to mitigate algorithmic bias—aligning technical innovation with social good

Android App Development

Creator & Lead Developer, SnapPrice (Android App)

2024 – Present

- **Korea Code Fair Semi-Finalist (Top 70):** Recognized for technical innovation among a nationwide pool of student developers and successfully published to the Google Play Store after a rigorous closed-testing phase, with live availability in South Korea and the United States.
- **Full-Stack Implementation:** Built the native Android interface (Java/Kotlin) utilizing **Google ML Kit** for barcode OCR and **Firebase Firestore** for real-time data synchronization.
- **Performance Optimization:** Achieved an average predictive accuracy of **81.5%** (peaking at **92.0%** for stable product categories) by implementing a "Buy vs. Wait" decision algorithm to empower rational consumer consumption.

Android App Development

Creator & Lead Developer, Curiosity Creators Lesson Creator App (Android App)

2025 – Present

- **AI-Powered Pedagogical Platform:** Designed and developed a classroom management and STEM lesson-planning utility utilizing Google Gemini to generate customized curricula and dynamically orchestrate team grouping based on individual performance analytics.
- **Modern Native Development & OCR:** Engineered the app entirely in Kotlin with Jetpack Compose using the MVVM architecture and Room Database; integrated Google ML Kit OCR to instantly scan and parse physical evaluation rubrics into digital student profiles.
- **Algorithmic Learner Profiling:** Devised the Curiosity Creators Score (CCS) heuristic—a weighted formula analyzing student performance across 6 cognitive dimensions (Planning, Iteration, Independence, Metacognition, Mastery, and Dynamics)—to assign targeted collaborative roles (e.g., Manager) to address specific learner deficits.

Research Experience

Lead Researcher, Independent Research

2025

- **Project:** *"A Hybrid LSTM-LLM Framework for 10-Year Treasury Yield Forecasting"*.
- Engineered a novel predictive architecture integrating **Long Short-Term Memory (LSTM)** networks with **LLM-generated sentiment analysis** (Gemini 1.5 Flash) to forecast complex macroeconomic indicators.
- Outperformed traditional econometric models by achieving a **17.4% reduction in RMSE** and **24.6% reduction in MAE**, proving that unstructured news data serves as a critical "leading indicator" compared to lagging metrics like CPI.
- Validated results through the **Diebold-Mariano Test ($p < 0.0001$)**, confirming that the integration of qualitative sentiment provides a statistically significant advantage in financial modeling.

Research Experience

Lead Researcher, Korea Science and Engineering Fair (KSEF)

2025

- **Project:** *"Overcoming the Persistence Trap in Stock Prediction: A Multi-Modal Approach Using Transformers and LLMs"*.
- Developed a novel Multi-Modal AI architecture combining **Transformers** for time-series analysis and **LLMs (Gemini 2.5 Pro)** for financial sentiment analysis.
- Solved the "Persistence Trap" (structural time-lag) in deep learning models; successfully reduced predictive autocorrelation from **0.9998 to 0.1575**, proving that news data triggers independent model inference rather than simple pattern copying.
- Conducted rigorous statistical verification (Granger Causality Test) to validate the **Efficient Market Hypothesis**, demonstrating the limits of algorithmic trading in high-efficiency markets.

Curiosity Creators

Director of Korea Chapter

10/2024 - Present

- **Engineered proprietary STEM educational software** to gamify learning, deploying the tool to simplify complex technical concepts for young students across the network.
- Direct operations contributing to a global impact of **1,000+ students** and **10+ major events**, actively managing partnerships with **6 external organizations** to expand outreach.
- Coordinate cross-chapter logistics, collaborating with international branches to organize joint workshops and standardize educational curriculum for wider accessibility.

KIS Makers Club

Club Leader / President

08/2024 - Present

- **Orchestrated a collaborative engineering seminar with visiting MIT students**, facilitating a "Real-World Problem Solving" workshop for **30+ peers** grounded in creative design thinking.
- Direct operations for a **50-member** organization, personally designing and executing the educational curriculum for weekly workshops on 3D printing and digital design.

- Mentor students through the **engineering design process**, guiding them from abstract ideation and troubleshooting to final hardware prototyping.

KIS Chess Club*Member*

08/2024 - Present

KIS Badminton Club*Member*

08/2024 - Present

SKILLS & INTERESTS

- **Technical:** Java (Android Studio), Python (TensorFlow, Keras, Pandas), SQL, R (Basic), Firebase, Google ML Kit, JavaScript
- **Data & AI:** LSTM/RNN Architectures, Transformers, LLM Integration, Time-Series Forecasting, Statistical Analysis
- **Languages:** English (Native), Korean (Native), Spanish (Advanced)
- **Athletics:** Figure Skating, Badminton